Duke | SANFORD SCHOOL *of* PUBLIC POLICY

PUBPOL813 Midterm Review

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Question 1

Consider a polynomial regression model of the following form:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2$$

Calculate, in terms of the parameters β_1, β_2, and β_3, what the estimated change in y would be when x changes from 2 to 3.

Question 2

Say you estimate a probit regression with only one dependent variable, y, two independent variables, x_1 and x_2, and the interaction between them, to obtain the parameters $\alpha_0, \alpha_1, \alpha_2$, and α_3. The z-score is then given by the equation $\alpha_0 + \alpha_1 x_1 + \alpha_2 x_2 + \alpha_3 x_1 x_2$. Calculate, **in terms of the normal CDF $\Phi()$ and the given parameters**, the change in the probability that $y = 1$ when x_1 remains constant at a value of 2 and x_2 changes from -4 to -3.

Question 3

Your colleague tells you about a recent RCT they worked on which came up with null results; they also explain that they used a sample size of 25 in the control group and 25 in the treatment group. Explain in words why you might be skeptical of these results.

Question 4

You are helping with an RCT that is taking place in 50 towns and which gives some households in each town access to public transportation vouchers. You run the following Stata command to try to determine the effect of the program on gas expenditures:

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reg gasexp treatment, robust cluster(town)
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Explain (in words) what the phrases after the comma do and why you should include them.

Question 5

You run the regression from question 4, but now with the logarithm of gas expenditures as the dependent variable. Stata reports that the coefficient on the treatment variable is -0.326, and the associated p-value is 0.002. Interpret these results.

Question 6

Considering still the transportation voucher question, suppose you now want to see if, **at the town level**, treatment has some impact on road safety. You thus regress the number of traffic accidents (per capita) in the town on the proportion of households in the town which received the vouchers, along with a variable representing the amount of money the town spent on road maintenance in the last year. Stata reports that the coefficient on the proportion of treatment households variable is -2.79, with an associated p-value of 0.12. Interpret and reflect on these results.

Question 7

As a final test of the effects of the transportation voucher, and **returning to the household level**, you decide to try to determine, using a probit model, what the effect of receiving the vouchers is on the probability of the household experienceing a car accident in the month after first receiving the voucher. You find that the estimated constant and slope coefficients associated with the regression (which includes no control variables) are 1.26 and -0.08, respectively. Interpret the slope coefficient and calculate the estimated change in the probability of a household experiencing an accident when receiving a voucher.

Question 8

You are designing an RCT that provides free supplementary textbooks to some classrooms across several school districts in a large, densely populated city. Discuss how the three threats to randomized experiments we talked about in class might be relevant to this study.